VTOL Pop-Up Repeaters for Amateur Radio and Environmental Design Log

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Course: URMP

Introduction

Welcome to Project UWU: Unmanned Wide-area Uplink:

This project focuses on designing and building a delta wing UAV that can carry a payload of 250 grams or more, including sensors and measurement tools. The drone will be built for long-endurance flights, with a target duration between 1 and 4 hours. All parts, including the frame, ESCs, motors, battery, GPS, and flight controller, will be selected for low cost and high efficiency.

The drone will fly primarily using GPS-based autonomous navigation, with manual override available through remote control. It should be easy for anyone to use. The goal is to create a stable, efficient, and affordable platform for collecting data in the field.

Attributes	
Dual motor	Yes
1-4 hr flight time	Questionable, maybe 1-2 hours
Size limit?	No
Delta Wing Style airframe	Yes
250g+ payload	Yes
Flight controller	Yes
Material	Undetermined, foamboard for gen 1
GPS	Yes
Lands and Launches vertically	Yes
Battery Size?	Undetermined

Make my drone legal

No one cares we are recreational, ok may need to register with the FAA since this drone will be way heavier than 250g

Choosing Frame Specs

We are planning a twin motor delta wing design with VTOL capability. Target wingspan is 1000mm, optimized for efficient gliding and long-duration cruise. Materials may include EPP foam or laser-cut foam board with carbon fiber spars for strength. Below is a general table on how to design such a drone.

Table of rules for VTOL Capability			
Category	Rule of Thumb	Explanation	Links
Thrust-to-weight	≥2:1 total thrust	Total vertical motor thrust must be at least 2× total weight to handle safe takeoff, hover, and basic maneuvering with margin	Thrust to weight ratio - ArduPlane - ArduPilot Discourse https://www.reddit.com/r/Multicopter/comments/56gt62/where_does_the_21_rule_of_thumb_on_thrusttoweight/
Motor sizing	1 motor per 500–1000g (for lift motors)	Balance motor weight and thrust. For endurance, use fewer, larger motors instead of many small ones	https://www.reddit.com/r/fpv/comments/12buu69/what_consumes_more_power_hovering_or_chill_fly/ https://ntrs.nasa.gov/api/citations/20180006683/downloads/20180006683.pdf?utm_source=chatgpt.com
Prop size	Larger, slower props = more efficient	10–13" props on 4S–6S systems are common; low pitch helps for stable VTOL and efficient hover	Props sized typically used on 5ft super Cub planes for extreme long flight times
Hover power draw	~60–80% of full throttle current	Plan for 30-40A total current draw during hover for ~2kg drones; don't size ESCs too tight.	Based on using 2814 1000KV motors with 10-13 inch props
Hover time vs. cruise	Hover burns 4–6 times more power	Minimize hover time - just long enough for takeoff/landing. Plan to spend 90%+ of flight in fixed-wing mode	Ask Ed are we hovering or gliding in a circle. We are gliding in a circle

Battery sizing	Use Li-ion for endurance, LiPo for power	Li-ion packs offer higher energy density, but lower current tolerance	Li-Ion Vs. Lipo Battery For Long-Range Flights | Grepow
Center of gravity (CG)	At or slightly ahead of wing's aerodynamic center(For delta planes)	Crucial for smooth transition between hover and forward flight; balance battery, motors, payloads carefully	Typical of Delta Type aicraft
Flight controller	Must support fixed wing aircraft and hovering	ArduPilot and PX4 are top choices; support vertical-to-fixed transitions, auto missions, and fallback modes	Ask Ed
ESCs	30–60A with braking, BLHeli_S or fixed-wing ESCs	Pick reliable, smooth throttle ESCs that support brake-on-stop for folding props (reduces drag in cruise)	Amazon.com: 45A ESC 2S-6S Brushless ESC LANRC Lipo BLHeli_S Electronic Speed Controller Support D-Shot 150/300/600 for FPV Racing Drone Multicopter Quadcopter 2PCS : Toys & Games
Airframe weight budget	Frame + motors + ESCs should not exceed 35% of the total drone	Save weight for battery and payload - keep structural and electronic components efficient	kcl.digimat.in Payload as a % of AUW | DroneVibes - Drones, UAV's, Multirotor, Professional, Aerial Photography, FPV, Drone Racing

We also need some kind of air foil to increase efficiency. I usually use Airfoil Tools website to find a reasonably sized airfoil for normal rc planes. But this is different, I need some way to calculate the appropriate airfoil I need, I need to design it in such a way the plane should fly mostly horizontally and climb to even out gravity. I don't know how to do that so I should speak with a specialist like Peter.

Since the payload is fairly heavy and we need VTOL for easier landings, I think we need to size the airframe properly. A 1200mm wingspan seems like a good choice.

We are building a wing similar to the Eflite X-VERT VTOL and the XK/WLTOYS x520

Choosing a Motor and Propeller(Prop)

On my Carl Goldberg Super Cub I'm using a 4120 710KV motor, it feels safe but slightly underpowered on 4s. So I think we should choose a higher KV to ensure we have enough RPM for more power. This aircraft is also incredibly heavy, roughly 8 lbs.

The Delta Wing Drone will probably weigh less than 2kg (4.4lbs) so we can get away with lighter materials.

Motors: Dual 2814 900KV-1000K motors (or similar) for efficient hover and cruise. I like what I call the Mitoot Style Motors, Motors with orange/gold top and bottom and silver can. I know these are very strong for the price.
Props: 10x5 or 11x5 or 10x6 slow-fly props - folding props maybe??
Each motor must produce at least 1.2 kg of thrust for vertical takeoff at ~1.6–1.8 kg total weight

D2836-850KV

Vendor	Motor	KV rating	Battery Cell Rating (Lipo)/Max thrust	Weight(Grams)	Total Cost for 2 motors (USD)
Amazon	2814	1000Kv	2-4s/unknown	52g	42
Amazon	FLASH HOBBY D2830	1000Kv	2-4s/890g	52g	38
Amazon	FLASH HOBBY D3542	1000Kv	2-5s/1260g	130g	46
Amazon	FLASH HOBBY D3542	1450Kv	2-5s/1260g	130g	46
Amazon	FLASH HOBBY D2830EVO	1000Kv	2-4s/930g	68.1g	40
	~~FlashHobby D4260 EVO~~	600kv	5-6s/2000g		
Amazon	D3548	790kv	3-5s/1650	156g	40
Amazon	D2836	1100Kv	2-4s/1130g	72g	38

Amazon	D2836	850KV	2-4s/850g	72g	38

Deep Dive into motor research

D2830 EVO Motor Test Data:

MOTOR TECHNICAL DATA:									
Item NO.	Volts(V)	Prop	Throttle(%)	CURRENT (A)	Power(W)	Thrust (g)	PRM	Efficiency (g/W)	Operating temperature(℃)
D2830-750KV(EVO)	12.65	APC 9x6	40%	1.89	23.9	231	4501	9.6	36.7℃
	12.63		50%	2.51	31.7	285	5014	9.0	
	12.61		60%	3.12	39.3	328	5463	8.3	
	12.59		70%	3.71	46.7	369	5836	7.9	
	12.53		80%	4.80	60.1	402	6358	6.7	
	12.47		90%	6.23	77.7	525	7020	6.8	
	12.42		100%	7.08	87.9	580	7278	6.6	
	16.69	APC 8x6	40%	2.46	41.1	280	6214	6.8	37.6℃
	16.66		50%	3.19	53.1	314	6856	5.9	
	16.63		60%	3.91	65.0	380	7387	5.8	
	16.60		70%	4.62	76.7	432	7855	5.6	
	16.54		80%	5.73	94.8	511	8420	5.4	
	16.46		90%	7.66	126.1	609	9226	4.8	
	16.40		100%	8.49	139.2	648	9583	4.7	
	16.69	APC 9x6	40%	2.74	45.7	346	5546	7.6	39.5℃
	16.66		50%	3.62	60.3	394	6234	6.5	
	16.61		60%	4.61	76.6	447	6693	5.8	
	16.56		70%	5.70	94.4	580	7261	6.1	
	16.48		80%	7.38	121.6	692	7953	5.7	
	16.35		90%	9.75	159.4	840	8672	5.3	
	16.27		100%	10.78	175.4	897	8974	5.1	
D2830-850KV(EVO)	12.48	APC 8x6	40%	2.04	25.5	218	5286	8.6	36.8℃
	12.46		50%	2.64	32.9	259	5857	7.9	
	12.44		60%	3.22	40.1	296	6301	7.4	
	12.41		70%	3.76	46.7	327	6702	7.0	
	12.39		80%	4.40	54.5	363	7142	6.7	
	12.32		90%	5.83	71.8	441	7862	6.1	
	12.29		100%	6.64	81.6	479	8184	5.9	
	12.56	10x5	40%	2.61	32.8	272	4400	8.3	38.3℃
	12.54		50%	3.54	44.4	346	4925	7.8	
	12.52		60%	4.44	55.6	412	5380	7.4	
	12.49		70%	5.70	71.2	494	5881	6.9	
	12.45		80%	7.57	94.2	604	6546	6.4	
	12.40		90%	10.09	125.1	741	7155	5.9	
	12.37		100%	11.27	139.4	795	7436	5.7	
D2830-850KV(EVO)	12.57	APC 11x7	40%	2.94	37.0	283	3742	7.7	39.3℃
	12.55		50%	4.02	50.5	362	4197	7.2	
	12.51		60%	5.54	69.3	464	4706	6.7	
	12.47		70%	7.88	98.3	591	5329	6.0	
	12.39		80%	10.79	133.7	725	5864	5.4	
	12.33		90%	14.18	174.8	842	6319	4.8	
	12.29		100%	15.87	195.0	876	6490	4.5	
	16.63	APC 8x6	40%	2.98	49.6	319	6624	6.4	37.4℃
	16.59		50%	3.91	64.9	384	7348	5.9	
	16.55		60%	4.80	79.4	443	7888	5.6	
	16.51		70%	5.72	94.4	494	8371	5.2	
	16.44		80%	7.14	117.4	574	9054	4.9	
	16.30		90%	9.43	153.7	686	9874	4.5	
	16.21		100%	10.67	173.0	733	10277	4.2	
D2830-1000KV(EVO)	12.38	APC 9x6	40%	3.70	45.8	301	5502	6.6	39.4℃
	12.32		50%	4.89	60.2	366	6150	6.1	
	12.25		60%	6.09	74.6	407	6663	5.5	

	12.13		70%	7.39	89.6	516	7117	5.8	
	11.96		80%	9.38	112.2	616	7739	5.5	
	11.83		90%	12.19	144.2	762	8458	5.3	
	11.76		100%	13.72	161.3	820	8765	5.1	
	12.31	10x5	40%	3.93	48.4	356	4997	7.4	40.4℃
	12.22		50%	5.30	64.8	446	5585	6.9	
	12.10		60%	6.69	80.9	526	6056	6.5	
	11.91		70%	8.74	104.1	632	6611	6.1	
	11.84		80%	11.38	134.7	761	7272	5.6	
	11.68		90%	14.85	173.4	902	7876	5.2	
	11.70		100%	16.80	196.6	953	8199	4.8	
	16.53	APC 8x6	40%	4.73	78.2	381	7638	4.9	40.3℃
	16.42		50%	6.22	102.1	471	8375	4.6	
	16.29		60%	7.65	124.6	544	8960	4.4	
	16.12		70%	9.06	146.0	605	9492	4.1	
	16.08		80%	11.72	188.5	713	10301	3.8	
	15.89		90%	15.55	247.1	855	11203	3.5	
	15.91		100%	17.79	283.0	930	11581	3.3	
D2830-1300KV(EVO)	12.08	APC9x6	40%	5.94	71.8	372	6336	5.2	41.3℃
	11.98		50%	7.98	95.6	510	7056	5.3	
	11.85		60%	10.09	119.6	605	7638	5.1	
	11.79		70%	12.54	147.8	720	8243	4.9	
	11.67		80%	16.28	190.0	880	8982	4.6	
	11.48		90%	21.06	241.8	1034	9701	4.3	
	11.53		100%	23.74	273.7	1108	10090	4.0	
	12.23	APC10x5	40%	6.41	78.4	490	5784	6.2	42.6℃
	12.07		50%	8.85	106.8	585	6473	5.5	
	11.84		60%	11.36	134.5	717	6993	5.3	
	11.83		70%	15.22	180.1	876	7725	4.9	
	11.61		80%	19.63	227.9	1013	8352	4.4	
	11.42		90%	25.31	289.0	1156	8951	4.0	
	11.45		100%	28.33	324.4	1228	9237	3.8	
Notes: The test condition oftemperature is motor surface temperature in 100% throttle while the motor run 15 s.environment temperature 26℃									

D2836 EVO Motor TEST DATA:

MOTOR TECHNICAL DATA:									
Item NO.	Volts(V)	Prop	Throttle(%)	CURRENT (A)	Power(W)	Thrust (g)	PRM	Efficiency (g/W)	Operating temperature(℃)
D2836-750KV(EVO)	12.46	APC 9x6	40%	1.95	24.3	220	5441	9.1	30.8℃
	12.45		50%	2.44	30.4	260	5932	8.6	
	12.44		60%	2.83	35.2	288	6341	8.2	
	12.43		70%	3.21	39.9	318	6639	8.0	
	12.41		80%	3.61	44.8	346	6929	7.7	
	12.39		90%	4.77	59.1	424	7666	7.2	
	12.38		100%	4.88	60.4	425	7766	7.0	
	16.59	APC 9x6	40%	3.44	57.1	414	5962	7.2	39.5℃
	16.56		50%	4.52	74.9	498	6612	6.7	
	16.53		60%	5.55	91.7	569	7107	6.2	
	16.50		70%	6.65	109.7	650	7573	5.9	
	16.44		80%	8.54	140.4	764	8323	5.4	
	16.33		90%	11.60	189.4	953	9182	5.0	
	16.31		100%	11.95	194.9	966	9260	5.0	
D2836-850KV(EVO)	12.35	APC 12x6	40%	2.80	34.6	265	3552	7.7	38.8℃
	12.31		50%	4.46	54.9	350	4030	6.4	
	12.29		60%	5.61	68.9	454	4604	6.6	
	12.26		70%	7.12	87.3	583	5217	6.7	
	12.21		80%	9.44	115.3	727	5727	6.3	
	12.14		90%	12.95	157.2	825	6296	5.2	
	12.13		100%	13.19	160.0	827	6316	5.2	
	16.59	10x5	40%	3.46	57.4	387	5102	6.7	39.3℃
	16.56		50%	4.98	82.5	494	5745	6.0	
	16.52		60%	6.47	106.9	611	6429	5.7	
	16.44		70%	8.51	139.9	736	7077	5.3	
	16.34		80%	11.12	181.7	875	7760	4.8	
	16.19		90%	14.66	237.3	1026	8409	4.3	
	16.18		100%	15.11	244.5	1028	8466	4.2	
	16.56	APC 11x7	40%	4.06	67.2	383	4217	5.7	39℃
	16.51		50%	6.02	99.4	497	4794	5.0	
	16.45		60%	8.16	134.2	630	5444	4.7	
	16.31		70%	11.64	189.8	738	5944	3.9	
	16.19		80%	14.58	236.1	859	6394	3.6	
	15.95		90%	19.66	313.6	1016	6826	3.2	
	15.91		100%	19.79	314.9	1011	6827	3.2	
D2836-1100KV (EVO)	12.38	APC 9x6	40%	5.76	71.3	438	6409	6.1	40.3℃
	12.32		50%	7.64	94.1	561	7095	6.0	
	12.25		60%	9.51	116.5	661	7642	5.7	
	12.19		70%	11.46	139.7	735	8163	5.3	
	12.03		80%	14.81	178.2	867	8907	4.9	
	11.71		90%	19.42	227.4	998	9666	4.4	
	11.66		100%	19.95	232.6	1008	9725	4.3	
	16.26	APC8x6	40%	6.71	109.1	509	8519	4.7	41.2℃
	16.19		50%	8.93	144.6	615	9367	4.3	
	16.12		60%	11.02	177.6	691	10035	3.9	
	16.04		70%	13.13	210.6	783	10635	3.7	
	15.78		80%	16.62	262.3	920	11320	3.5	
	15.47		90%	21.54	333.2	1049	12247	3.1	
	15.32		100%	24.46	374.7	1138	12580	3.0	
	16.26	APC 9x6	40%	7.72	125.5	667	7637	5.3	42.4℃
	16.17		50%	10.41	168.3	794	8453	4.7	
	16.07		60%	13.03	209.4	912	9119	4.4	
	15.84		70%	15.92	252.2	956	9719	3.8	
	15.63		80%	20.92	327.0	1210	10617	3.7	
	15.35		90%	27.22	417.8	1456	11466	3.5	

	15.27		100%	30.30	462.7	1623	11804	3.5	
D2836-1450KV(EVO)	12.25	APC9x6	40%	9.81	120.2	618	7540	5.1	43.5℃
	12.04		50%	13.06	157.2	743	8280	4.7	
	11.91		60%	16.21	193.1	879	8883	4.6	
	11.82		70%	19.69	232.7	995	9475	4.3	
	11.63		80%	24.92	289.8	1178	10239	4.1	
	11.60		90%	32.92	381.9	1429	11194	3.7	
	11.58		100%	37.35	432.5	1535	11602	3.5	
	15.97	APC8x4	40%	12.13	193.7	704	10860	3.6	44.2℃
	15.82		50%	16.29	257.7	847	11467	3.3	
	15.67		60%	19.77	309.8	945	12423	3.1	
	15.70		70%	25.24	396.3	1108	13360	2.8	
	15.48		80%	32.96	510.2	1292	14290	2.5	
	15.26		90%	43.38	662.0	1497	14668	2.3	
	15.19		100%	48.69	739.6	1582	16579	2.1	
Notes: The test condition oftemperature is motor surface temperature in 100% throttle while the motor run 15 s.environment temperature 26℃									

D2826 EVO Motor Test Data:

MOTOR TECHNICAL DATA:									
Item NO.	Volts(V)	Prop	Throttle(%)	CURRENT (A)	Power(W)	Thrust (g)	PRM	Efficiency (g/W)	Operating temperature(℃)
D2826-930KV(EVO)	12.56	APC 9x6	40%	2.66	33.4	266	4413	8.0	36.7℃
	12.54		50%	3.66	45.9	344	4972	7.5	
	12.52		60%	4.78	59.8	421	5483	7.0	
	12.48		70%	6.70	83.6	522	6153	6.2	
	12.42		80%	9.02	112.0	655	6808	5.9	
	12.37		90%	11.79	145.8	781	7409	5.4	
	12.34		100%	13.20	162.9	840	7647	5.2	
	12.48	APC 9x6	40%	2.53	31.6	225	4806	7.1	36.3℃
	12.46		50%	3.40	42.4	288	5418	6.8	
	12.44		60%	4.29	53.4	360	5910	6.7	
	12.41		70%	5.55	68.9	378	6509	5.5	
	12.37		80%	7.41	91.7	529	7181	5.8	
	12.33		90%	9.71	119.7	650	7881	5.4	
	12.31		100%	10.89	134.1	688	8189	5.1	
D2826-1000KV(EVO)	12.62	APC 9x6	40%	2.99	37.7	280	5084	7.4	36.2℃
	12.55		50%	3.97	49.8	314	5726	6.3	
	12.47		60%	5.03	62.7	380	6250	6.1	
	12.27		70%	6.55	80.4	432	6824	5.4	
	12.14		80%	8.56	103.9	511	7476	4.9	
	12.06		90%	11.15	134.5	609	8175	4.5	
	12.00		100%	12.50	150.0	648	8416	4.3	
	16.63	APC 8x6	40%	3.87	64.4	335	7104	5.2	37.5℃
	16.55		50%	5.22	86.4	413	7858	4.8	
	16.45		60%	6.52	107.3	489	8523	4.6	
	16.33		70%	8.07	131.8	566	9113	4.3	
	16.10		80%	10.50	169.1	674	9922	4.0	
	16.01		90%	13.95	223.3	805	10742	3.6	
	15.94		100%	15.67	249.8	847	11100	3.4	
	16.56	APC 9x6	40%	4.16	68.9	397	6283	5.8	38.8℃
	16.46		50%	5.70	93.8	493	6992	5.3	
	16.31		60%	7.44	121.3	605	7639	5.0	
	16.21		70%	9.94	161.1	758	8398	4.7	
	16.00		80%	12.90	206.4	917	9156	4.4	
	15.88		90%	16.62	263.9	1050	9886	4.0	
	15.85		100%	18.43	292.1	1111	10141	3.8	
D2826-1450KV(EVO)	12.49	7x5	40%	6.44	80.4	340	7639	4.2	38.5℃
	12.46		50%	8.68	108.2	423	8486	3.9	
	12.41		60%	11.01	136.6	500	9187	3.7	
	12.35		70%	13.59	167.8	584	9883	3.5	
	12.25		80%	17.86	218.8	711	10820	3.2	
	12.13		90%	23.15	280.8	835	11707	3.0	
	12.07		100%	25.83	311.8	887	12081	2.8	
	12.49	APC 8x6	40%	6.45	80.6	393	7565	4.9	39.2℃
	12.44		50%	8.77	109.1	480	8388	4.4	
	12.39		60%	11.19	138.6	565	9024	4.1	
	12.34		70%	13.85	170.9	649	9708	3.8	
	12.22		80%	18.49	225.9	778	10583	3.4	
	12.09		90%	24.27	293.4	909	11409	3.1	
	12.00		100%	27.09	325.1	956	11673	2.9	
	12.46	APC 9x6	40%	7.04	87.7	416	6668	4.7	40.1℃
	12.40		50%	9.78	121.3	592	7455	4.9	
	12.33		60%	13.05	160.9	708	8202	4.4	
	12.24		70%	16.67	204.0	855	8912	4.2	
	12.11		80%	21.72	263.0	1017	9704	3.9	
	11.92		90%	28.11	335.1	1194	10381	3.6	

	11.85		100%	30.97	367.0	1256	10645	3.4	
D2826-2200KV(EVO)	12.41	APC9x6	40%	8.53	105.9	328	17323	3.1	39.6℃
	12.37		50%	10.54	130.4	387	18730	3.0	
	12.33		60%	12.34	152.2	442	19866	2.9	
	12.29		70%	13.86	170.3	500	20750	2.9	
	12.26		80%	15.20	186.4	538	21420	2.9	
	12.19		90%	17.35	211.5	583	22561	2.8	
	12.14		100%	19.42	235.8	653	23470	2.8	
	12.34	EP6050	40%	10.35	127.7	448	14953	3.5	40.9℃
	12.28		50%	13.11	161.0	547	16434	3.4	
	12.22		60%	15.48	189.2	626	17576	3.3	
	12.17		70%	17.58	213.9	698	18557	3.3	
	12.11		80%	19.64	237.8	772	19451	3.2	
	11.99		90%	23.63	283.3	896	20856	3.2	
	11.92		100%	26.13	311.5	963	21658	3.1	
Notes: The test condition oftemperature is motor surface temperature in 100% throttle while the motor run 15 s.environment temperature 26℃									

D3530 EVO RC Brushless Motor Test Datas

MOTOR TECHNICAL DATA:									
Item NO.	Volts(V)	Prop	Throttle(%)	CURRENT (A)	Power(W)	Thrust (g)	PRM	Efficiency (g/W)	Operating temperature(℃)
D3530-1100KV (EVO)	12.45	7x5	40%	3.10	38.6	221	6162	5.7	37.2℃
	12.43		50%	4.21	52.3	284	6874	5.4	
	12.41		60%	5.30	65.8	311	7441	4.7	
	12.38		70%	6.63	82.1	373	8097	4.5	
	12.34		80%	8.83	109.0	436	8941	4.0	
	12.28		90%	11.62	142.7	519	9862	3.6	
	12.26		100%	13.00	159.4	565	10156	3.5	
	12.82	APC8x6	40%	3.33	42.7	264	6299	6.2	38.3℃
	12.75		50%	4.53	57.8	331	7043	5.7	
	12.60		60%	5.80	73.1	393	7639	5.4	
	12.45		70%	7.36	91.6	446	8323	4.9	
	12.43		80%	9.93	123.4	555	9178	4.5	
	12.40		90%	13.29	164.8	670	10060	4.1	
	12.33		100%	14.75	181.9	715	10347	3.9	
	16.86	7x5	40%	4.65	78.4	352	7799	4.5	39.1℃
	16.75		50%	6.45	108.0	433	8691	4.0	
	16.70		60%	8.35	139.4	513	9496	3.7	
	16.50		70%	10.82	178.5	627	10363	3.5	
	16.45		80%	14.86	244.4	753	11467	3.1	
	16.32		90%	19.51	318.4	929	12467	2.9	
	16.22		100%	21.52	349.1	958	12764	2.7	
	16.77	APC8x6	40%	4.71	79.0	390	7660	4.9	39.6℃
	16.70		50%	6.56	109.6	477	8590	4.4	
	16.65		60%	8.69	144.7	582	9370	4.0	
	16.50		70%	11.44	188.8	695	10227	3.7	
	16.48		80%	15.92	262.4	877	11264	3.3	
	16.43		90%	21.12	347.0	1021	12125	2.9	
	16.36		100%	23.27	380.7	1080	12349	2.8	
D3530-1400KV(EVO)	12.39	7x5	40%	5.52	68.4	296	7357	4.3	38.5℃
	12.35		50%	7.56	93.4	352	8238	3.8	
	12.31		60%	9.78	120.4	412	8985	3.4	
	12.27		70%	12.31	151.0	504	9766	3.3	
	12.18		80%	16.96	206.6	662	10790	3.2	
	12.07		90%	22.19	267.8	784	11772	2.9	
	12.01		100%	24.38	292.8	826	12111	2.8	
	12.86	APC8x6	40%	6.41	82.4	442	6553	5.4	39.3℃
	12.80		50%	9.09	116.4	607	7405	5.2	
	12.79		60%	12.93	165.4	756	8315	4.6	
	12.69		70%	17.57	223.0	919	9142	4.1	
	12.56		80%	23.26	292.1	1046	10014	3.6	
	12.38		90%	30.30	375.1	1284	10721	3.4	
	12.28		100%	32.63	400.7	1351	10891	3.4	
	12.38	APC9x6	40%	6.01	74.4	356	6418	4.8	39.8℃
	12.34		50%	8.49	104.8	538	7272	5.1	
	12.27		60%	11.96	146.7	693	8133	4.7	
	12.19		70%	16.14	196.7	861	8949	4.4	
	12.08		80%	21.74	262.6	1066	9773	4.1	
	11.93		90%	28.34	338.1	1195	10569	3.5	
	11.86		100%	30.83	365.6	1283	10790	3.5	
	16.28	APC8x6	40%	8.58	139.7	654	7679	4.7	40.7℃
	16.19		50%	12.66	205.0	836	8724	4.1	
	16.07		60%	18.34	294.7	1009	9730	3.4	
	15.84		70%	24.11	381.9	1241	10598	3.2	
	15.65		80%	31.61	494.7	1469	11341	3.0	
	15.47		90%	40.16	621.3	1650	12010	2.7	

	15.43		100%	42.97	663.0	1662	12070	2.5	
D3530-1700KV(EVO)	12.65	7x5	40%	9.17	116.0	430	8664	3.7	39.6℃
	12.60		50%	12.77	160.9	522	9630	3.2	
	12.52		60%	16.68	208.8	614	10457	2.9	
	12.40		70%	21.83	270.7	762	11455	2.8	
	12.23		80%	28.73	351.4	912	12538	2.6	
	11.95		90%	37.08	443.1	1100	13430	2.5	
	11.92		100%	40.81	486.5	1167	13725	2.4	
	12.68	APC8x6	40%	9.29	117.8	502	8524	4.3	40.3℃
	12.61		50%	13.07	164.8	620	9477	3.8	
	12.53		60%	17.41	218.1	737	10299	3.4	
	12.42		70%	23.26	288.9	894	11222	3.1	
	12.27		80%	31.38	385.0	1056	12150	2.7	
	12.07		90%	41.31	498.6	1207	12963	2.4	
	11.99		100%	44.51	533.7	1229	13224	2.3	
	16.57	APC7x4	40%	10.37	171.8	545	11871	3.2	40.3℃
	16.49		50%	13.85	228.4	716	13091	3.1	
	16.37		60%	17.67	289.3	869	14130	3.0	
	16.14		70%	22.83	368.5	1046	15259	2.8	
	15.92		80%	30.40	484.0	1255	16705	2.6	
	15.65		90%	39.43	617.1	1474	17890	2.4	
	15.64		100%	44.92	702.5	1577	18461	2.2	
Notes: The test condition oftemperature is motor surface temperature in 100% throttle while the motor run 15 s.environment temperature 26℃									

3536 Brushless Motor Test Datas:

MOTOR TECHNICAL DATA:									
Item NO.	Volts(V)	Prop	Throttle(%)	CURRENT (A)	Power(W)	Thrust (g)	PRM	Efficiency (g/W)	Operating temperature(℃)
3536-910KV(EVO)	12.78	7x5	40%	3.33	42.6	239	6309	5.6	44.1℃
	12.76		50%	4.23	54.0	292	6938	5.4	
	12.75		60%	5.07	64.6	336	7439	5.2	
	12.73		70%	5.78	73.6	356	7875	4.8	
	12.71		80%	6.77	86.0	413	8213	4.8	
	12.68		90%	8.66	109.8	504	9064	4.6	
	12.66		100%	9.85	124.7	543	9487	4.4	
	12.81	APC8x6	40%	3.42	43.8	249	6432	5.7	45.2℃
	12.75		50%	4.45	56.7	313	7050	5.5	
	12.61		60%	5.34	67.3	369	7541	5.5	
	12.45		70%	6.17	76.8	405	7964	5.3	
	12.42		80%	7.15	88.8	472	8375	5.3	
	12.38		90%	9.54	118.1	580	9252	4.9	
	12.32		100%	11.06	136.3	650	9613	4.8	
	16.81	7x5	40%	4.78	80.4	369	7877	4.6	47.3℃
	16.74		50%	6.22	104.1	448	8654	4.3	
	16.70		60%	7.57	126.4	523	9311	4.1	
	16.56		70%	8.90	147.4	597	9825	4.1	
	16.48		80%	10.81	178.1	689	10509	3.9	
	16.44		90%	14.34	235.7	845	11604	3.6	
	16.41		100%	16.40	269.1	917	12056	3.4	
	16.72	APC8x6	40%	4.77	79.8	388	7795	4.9	48.3℃
	16.70		50%	6.40	106.9	489	8586	4.6	
	16.66		60%	7.70	128.3	558	9210	4.3	
	16.48		70%	9.13	150.5	641	9707	4.3	
	16.41		80%	11.36	186.4	732	10408	3.9	
	16.34		90%	15.29	249.8	883	11463	3.5	
	16.21		100%	17.54	284.3	992	11871	3.5	
	16.59	APC9x6	40%	5.35	88.8	520	6892	5.9	49.8℃
	16.55		50%	7.21	119.3	638	7618	5.3	
	16.51		60%	8.94	147.6	744	8256	5.0	
	16.47		70%	10.90	179.5	867	8811	4.8	
	16.39		80%	14.21	232.9	1051	9700	4.5	
	16.26		90%	19.10	310.6	1293	10615	4.2	
	16.21		100%	21.45	347.7	1393	10997	4.0	
3536-1000KV(EVO)	12.66	APC8x6	40%	4.61	58.4	323	7008	5.5	55.2℃
	12.60		50%	5.98	75.3	392	7739	5.2	
	12.53		60%	7.28	91.2	452	8294	5.0	
	12.50		70%	8.47	105.9	506	8745	4.8	
	12.44		80%	10.18	126.6	582	9266	4.6	
	12.40		90%	13.37	165.8	696	10230	4.2	
	12.35		100%	15.44	190.7	768	10640	4.0	
	12.87	APC10x5	40%	5.58	71.8	485	5855	6.8	56.4℃
	12.75		50%	7.65	97.5	596	6546	6.1	
	12.70		60%	9.70	123.2	720	7099	5.8	
	12.64		70%	12.11	153.1	837	7672	5.5	
	12.56		80%	16.22	203.7	1010	8488	5.0	
	12.31		90%	21.87	269.2	1254	9287	4.7	
	12.28		100%	24.44	300.1	1332	9646	4.4	
	16.44	7x5	40%	6.39	105.1	417	8597	4.0	52.1℃
	16.32		50%	8.39	136.9	520	9476	3.8	
	16.30		60%	10.31	168.1	604	10183	3.6	
	15.89		70%	12.23	194.3	670	10806	3.4	

	15.67		80%	15.19	238.0	837	11638	3.5	
	15.55		90%	20.07	312.1	1005	12837	3.2	
	15.46		100%	22.52	348.2	1100	13350	3.2	
3536-1250KV(EVO)	12.87	APC9x6	40%	8.29	106.7	580	7244	5.4	60.5℃
	12.82		50%	11.16	143.1	718	8043	5.0	
	12.77		60%	13.92	177.8	831	8702	4.7	
	12.72		70%	16.97	215.9	973	9301	4.5	
	12.62		80%	22.19	280.0	1164	10224	4.2	
	12.58		90%	29.53	371.5	1411	11156	3.8	
	12.50		100%	33.48	418.5	1547	11552	3.7	
	12.82	APC 10x5	40%	8.75	112.2	637	6691	5.7	63.1℃
	12.77		50%	12.04	153.8	812	7491	5.3	
	12.70		60%	15.58	197.9	958	8096	4.8	
	12.62		70%	19.61	247.5	1123	8792	4.5	
	12.50		80%	26.48	331.0	1386	9621	4.2	
	12.44		90%	34.87	433.8	1639	10450	3.8	
	12.34		100%	39.41	486.3	1739	10812	3.6	
	16.79	APC 10x5	40%	12.45	209.0	937	6691	4.5	69.6℃
	16.69		50%	17.56	293.1	1197	7491	4.1	
	16.58		60%	23.14	383.7	1421	8096	3.7	
	16.40		70%	30.35	497.7	1722	8792	3.5	
	16.28		80%	40.64	661.6	2061	9621	3.1	
	16.14		90%	52.79	852.0	2370	10450	2.8	
	16.04		100%	57.31	919.3	2507	10812	2.7	
3536-1450KV(EVO)	12.89	APC9x6	40%	11.14	143.6	682	7838	4.7	66.8℃
	12.82		50%	15.20	194.9	843	8711	4.3	
	12.76		60%	19.14	244.2	997	9446	4.1	
	12.68		70%	23.39	296.6	1150	10112	3.9	
	12.53		80%	31.06	389.2	1402	11054	3.6	
	12.34		90%	40.18	495.8	1669	11932	3.4	
	12.23		100%	45.45	555.9	1788	12301	3.2	
	12.68	APC11x7	40%	11.83	150.0	747	7177	5.0	65.6℃
	12.54		50%	16.31	204.5	940	7971	4.6	
	12.37		60%	20.86	258.0	1089	8624	4.2	
	12.28		70%	26.79	329.0	1314	9296	4.0	
	12.14		80%	35.38	429.5	1510	10184	3.5	
	11.86		90%	45.13	535.2	1787	10876	3.3	
	11.84		100%	51.45	609.2	1884	11260	3.1	

D3542 EVO Motor Test Data:

MOTOR TECHNICAL DATA:									
Item NO.	Volts(V)	Prop	Throttle(%)	CURRENT (A)	Power(W)	Thrust (g)	PRM	Efficiency (g/W)	Operating temperature(℃)
D3542-1000KV(EVO)	12.47	10x5	40%	7.29	90.9	591	6204	5.6	41.5℃
	12.43		50%	9.69	120.4	723	6871	5.4	
	12.39		60%	11.97	148.3	828	7409	5.2	
	12.35		70%	14.16	174.9	947	7865	4.8	
	12.29		80%	17.53	215.4	1088	8495	4.8	
	12.18		90%	23.58	287.2	1346	9342	4.6	
	12.13		100%	26.84	325.6	1451	9703	4.4	
	12.44	APC 11x7	40%	8.32	103.5	623	5289	6.0	43.8℃
	12.40		50%	11.38	141.1	771	5887	5.5	
	12.34		60%	14.87	183.5	914	6381	5.0	
	12.25		70%	19.84	243.0	1090	6994	4.5	
	12.13		80%	26.48	321.2	1304	7632	4.1	
	11.97		90%	35.49	424.8	1523	8269	3.6	
	11.87		100%	40.27	478.0	1615	8517	3.4	
	12.46	APC 12x6	40%	8.08	100.7	691	5470	6.9	44.3℃
	12.42		50%	11.10	137.9	829	6120	6.0	
	12.36		60%	14.23	175.9	1002	6622	5.7	
	12.29		70%	18.24	224.2	1196	7175	5.3	
	12.18		80%	24.08	293.3	1444	7868	4.9	
	12.04		90%	32.14	387.0	1736	8514	4.5	
	11.96		100%	36.70	438.9	1867	8805	4.3	
	12.45	APC 13x6.5	40%	8.88	110.6	737	4534	6.7	47.3℃
	12.38		50%	12.75	157.8	948	5121	6.0	
	12.29		60%	18.04	221.7	1204	5696	5.4	
	12.15		70%	25.77	313.1	1392	6325	4.4	
	11.98		80%	34.80	416.9	1777	6820	4.3	
	11.77		90%	44.94	528.9	2033	7267	3.8	
	11.65		100%	50.28	585.8	2136	7417	3.6	
D3542-1250KV(EVO)	12.48	APC9x6	40%	10.18	127.0	639	7631	5.0	44.2℃
	12.42		50%	13.52	167.9	784	8457	4.7	
	12.37		60%	16.59	205.2	909	9072	4.4	
	12.31		70%	19.45	239.4	983	9589	4.1	
	12.25		80%	23.55	288.5	1097	10238	3.8	
	12.10		90%	30.98	374.9	1395	11177	3.7	
	12.04		100%	34.89	420.1	1530	11619	3.6	
	12.47	APC10x6	40%	11.12	138.7	717	7114	5.2	48.4℃
	12.41		50%	14.94	185.4	891	7884	4.8	
	12.33		60%	18.72	230.8	1048	8451	4.5	
	12.27		70%	22.13	271.5	1160	8995	4.3	
	12.16		80%	28.18	342.7	1387	9695	4.0	
	12.00		90%	36.68	440.2	1646	10572	3.7	
	11.90		100%	41.92	498.8	1789	10923	3.6	
	16.34	APC10x6	40%	15.76	257.5	1079	8558	4.2	52.3℃
	16.24		50%	21.35	346.7	1331	9450	3.8	
	16.14		60%	26.91	434.3	1550	10140	3.6	
	16.02		70%	33.31	533.6	1788	10871	3.3	
	15.82		80%	43.24	684.1	2078	11807	3.0	
	15.47		90%	56.42	872.8	2438	12614	2.8	
	15.28		100%	62.94	961.7	2567	12919	2.7	
D3542-1400KV(EVO)	12.33	APC9x6	40%	12.92	159.3	757	8159	4.8	47.5℃
	12.26		50%	17.26	211.6	926	9036	4.4	
	12.19		60%	21.29	259.5	1085	9721	4.2	
	12.12		70%	25.24	305.9	1233	10239	4.0	
	12.02		80%	30.55	367.2	1408	10920	3.8	
	11.85		90%	39.43	467.2	1670	11878	3.6	

	11.74		100%	44.78	525.7	1815	12293	3.5	
	12.32	APC 10x5	40%	14.08	173.5	840	7573	4.8	55.1℃
	12.23		50%	18.97	232.0	1021	8382	4.4	
	12.15		60%	23.71	288.1	1196	9005	4.2	
	12.06		70%	28.34	341.8	1361	9529	4.0	
	11.93		80%	35.67	425.5	1593	10302	3.7	
	11.71		90%	46.11	539.9	1861	11112	3.4	
	11.57		100%	53.02	613.4	2036	11465	3.3	
Notes: The test condition oftemperature is motor surface temperature in 100% throttle while the motor run 15 s.environment temperature 26℃									

Choosing ESC

We need 50A esc with advanced programming features to tune for endurance. So we can just use any BLHeli based esc in our project.
Amazon.com: BLHeli RC Brushless ESC 12A 20A 30A 40A 50A 60A 80A with UBEC Motor Electric Speed Controller for Quadcopter Aircraft Model Fixed Wing Multi-axis DIY FPV RC Drone (20A) : Toys & Games

Or

Amazon.com: HGLRC Specter 90A 8S BLheli32 Single ESC High-Performance Professional ESC for FPV Racing Drone Freestyle (Specter 12S Distribution Board) : Toys & Games

Choosing a flight controller + sensors

We want PX4 based flight controller
Ask ED

Flight Controller	Notes
SpeedyBee F405 WING APP Fixed Wing Flight Controller	Limited by Non-native support for VTOL
Pixhawk 6C	Good option. A bit to new so info is hard to find. Expensive ~ 300
Pixhawk 4	Good option: Well documented for VTOL Expensive.
Pixhawk 2.4.8	~209
T1 VTOL FX-405 Flight Controller (with GPS & PMU)	Affordable, thats about it

We need GPS, barometer,

Choosing Controller electronics

I already have a RadioMaster Boxer Radio Controller. All we need is an ELRS Receiver. The longest range RX would be the RP3 Receiver from Radiomaster.
Amazon.com: RadioMaster RP3 ELRS Receiver 2.4GHz ExpressLRS Nano Receiver FPV RX Module Receiver Long Range Low Latency for FPV Micro Drone Racing Drone Quadcopter : Toys & Games

We also need a Module:
I found the Radiomaster Nomad, supports both 2.4Ghz and 900Mhz signals.

Freqncy band that we will dedicate
ELRS RX @ 915 MHz
Telemetry @ 433 MHz (to avoid clashing)
Video @ 5.8 GHz

Choosing a power source(ChatGpt)

For long-range fixed-wing drones, lithium-ion (Li-ion) batteries are preferred over lithium-polymer (LiPo) packs. Li-ion cells offer much higher energy density, allowing for significantly longer flight times at the cost of lower peak current output. Since fixed-wing drones typically cruise at low current draw, Li-ion is ideal for endurance-focused missions.

In this project, we use a 4S4P Li-ion pack (14.8V, ~16.8Ah) built from Molicel P42A or Sony VTC6 cells. This provides around 248 watt-hours of usable energy, supporting flight times of 2 hours or more depending on throttle and glide efficiency. Li-ion also has the advantage of lighter weight per watt-hour, which helps offset the heavier payload and VTOL motors.

LiPo may still be used for shorter missions, higher-power VTOL phases, or for test flights, but Li-ion is the better long-range solution.

I asked Ed to buy Sony VTC6 18650 Cells! The total battery weight

Additional GPS Navigation + Compass etc and possible FPV cameras

Since we are going to have 5-10 of these UAV, its crucial they all have accurate GPS systems for landing. We have 3 popular options, the M8, M9, and M10 series

What's GNSS?
GNSS stands for Global Navigation Satellite System, which is just a fancy way of saying "satellite systems that tell you where you are." GPS is the U.S. version, but other countries have their own—like GLONASS (Russia), Galileo (Europe), and BeiDou (China). Older GPS modules might only listen to one of these, but newer ones like the M9N and M10 can listen to all of them at once. That's called multi-GNSS. It's like having four sets of ears instead of one, so your drone can figure out its location faster and more accurately, even if the sky is cloudy or you're in a tricky spot.

What's Update Rate?
The update rate (measured in Hz) tells you how often the GPS sends a new position to your flight controller.

Model	GNSS Systems	Update Rate	Sensitivity	Compass	Power Draw	Firmware Compatibility	Pros	Cons	Cost Estimate
M8	GPS, GLONASS (sometimes Galileo, BeiDou)	Up to 10 Hz	Low in poor weather	Usually included	High (~50 mW)	Fully supported	Cheap, widely available	Poor in cloudy/urban conditions	$20–$35
M9N	GPS, GLONASS, Galileo, BeiDou	Up to 25 Hz	Good	Usually included (IST8310)	Moderate (~48 mW)	Fully supported	High update rate, reliable, mature support	Slightly higher power than M10	$50–$70
M10	GPS, GLONASS, Galileo, BeiDou	Max 10 Hz	Excellent (better cold start & weak signal performance)	Varies by module (many do not include)	Low (~25 mW)	Requires PX4 v1.14+ or ArduPilot 4.3+	Newer chip, better signal acquisition, lower power	Lower update rate, fewer modules include compass	$60–$90

*Highlighted is the one I chose

We can choose either the M9N or the M10. I'm going with the M10 because it uses about half the power and isn't much more expensive. While it has a lower update rate, I'm designing a cruiser, and 10 position updates per second is more than enough for my needs.

Videos on Pixhawk Setup
[(1/5) PixHawk Video Series - Simple initial setup, config and calibration]
[PixHawk/ArduCopter for Beginners (2023 series): 1. Flashing the PixHawk 6C and basic setup steps]

~~Solar Panels~~(Solars Panels are expensive and heavy)

Charging circuit CN3791

What does the tail cone do?

1. Improves Aerodynamic Stability
Extending the fuselage behind the elevator helps smooth out airflow around the tail.

It reduces turbulence that might affect the control surfaces and overall flight stability.

2. Shifts the Center of Pressure
A longer rear fuselage can increase the tail moment arm (distance from CG to tail), giving the elevator more leverage. That helps with pitch control and stability—especially useful in gliders, trainers, or slow flyers.

3. Protects Control Surfaces
It can shield the elevator from ground strikes during takeoff, landing, or crashes.

4. Structural or Aesthetic Purpose
Some models just replicate a full-scale design, which includes a longer tail section for aesthetic or aerodynamic continuity.

5. Improves efficiency
Smoother airflow reduces drag and improves glide ratio. In turn improves battery life up to 10%

Summary